Notes 4

1. How to Install and Remove Software Using APT

APT (Advanced Package Tool) is used in Linux to install, update, and remove software using the terminal.

Step 1: Open the Terminal

Open the terminal in Linux by pressing `Ctrl + Alt + T`.

Step 2: Update the Package List

Before installing software, update the package list.

```
sudo apt update
```

This command updates the list of available software packages.

Step 3: Install Software

To install software, use the `apt install` command.

Example 1: Install Tree

```
sudo apt install tree
```

The `tree` command displays files and folders in a tree structure.

Example 2: Install Htop

```
sudo apt install htop
```

Htop is used to view running processes and system performance.

Example 3: Install Curl

```
sudo apt install curl
```

Curl is used to transfer data from websites and servers.

Step 4: Remove Software

To remove installed software, use the `apt remove` command.

```
sudo apt remove tree
```

This removes the Tree program from Linux.

Step 5: Remove Software Completely

```
sudo apt purge tree
```

This removes the program and its system configuration files.

Step 6: Remove Unused Packages

```
sudo apt autoremove
```

This removes dependencies that are no longer needed.

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2. How to Create and Run a Shell Script

A shell script is a text file that contains Linux commands. It allows us to run multiple commands together.

Step 1: Create a Shell Script

Open the **Text Editor** in Linux and create a new file.

Save the file with a `.sh` extension.

For example:

`myscript.sh`

Step 2: Add the Shell Declaration

At the beginning of the file, type:

```
#!/bin/bash
```

This tells Linux to use the Bash shell to run the script.

Step 3: Add Your Code

Below the shell declaration, write the commands you want to run.

Example:

```
#!/bin/bash

echo "Hello Linux"
echo "Today's date is:"
date
echo "Current user:"
whoami
echo "Current directory:"
pwd
```

- `echo` displays a message.
- `date` shows the current date and time.
- `whoami` shows the current username.
- `pwd` shows the current directory.

After writing the code, save the file in the Text Editor.

Step 4: Run the Shell Script

Open the terminal in the folder where the script was saved.

First, give the script permission to run:

```
chmod +x myscript.sh
```

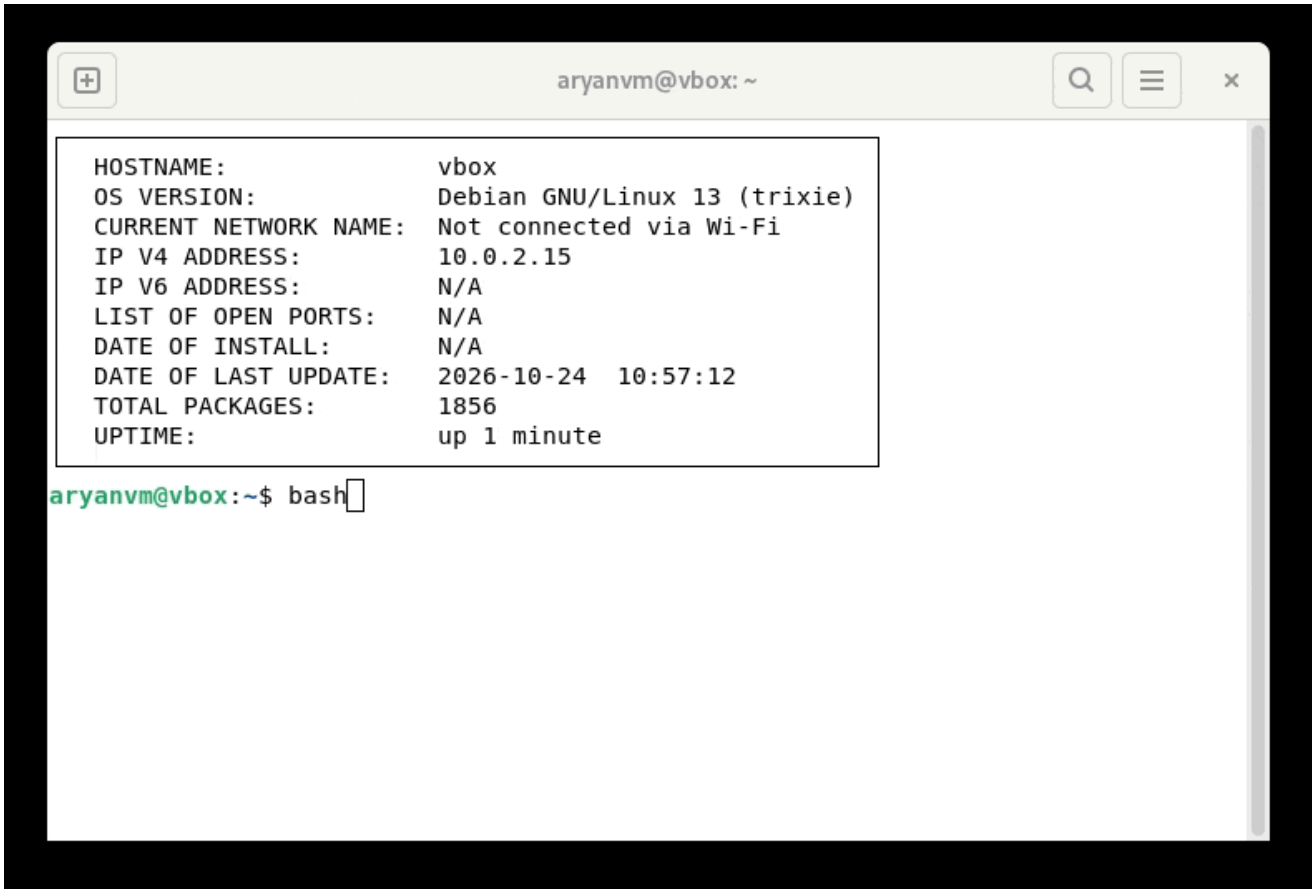
Then run the script:

```
./myscript.sh
```

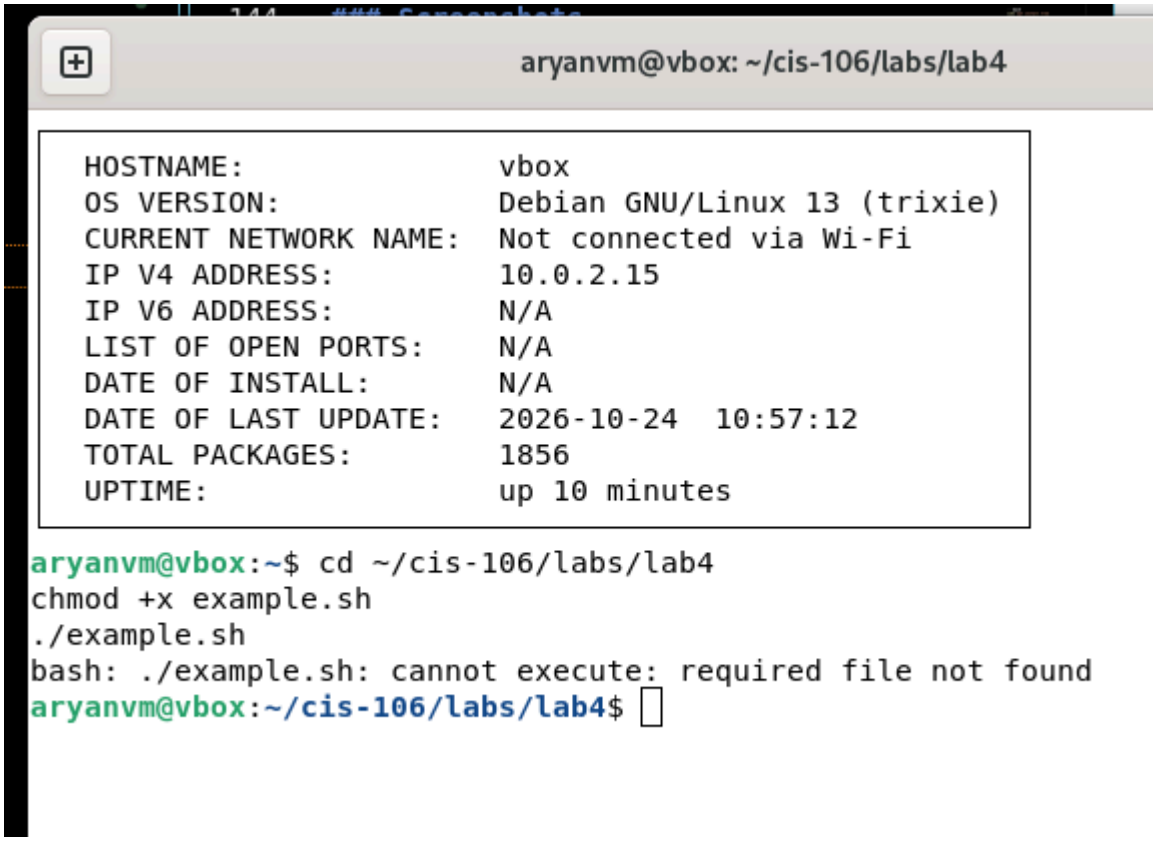
The terminal will display the message, current date, username, and directory.

Screenshots

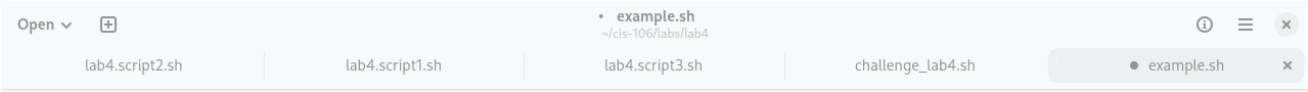
Screenshot 1: My shell script in the Text Editor



Screenshot 2: Giving execute permission in the terminal



Screenshot 3: Running the shell script


Open
example.sh
~/cis-106/labs/lab4
lab4.script2.sh
lab4.script1.sh
lab4.script3.sh
challenge_lab4.sh
example.sh